

Project Report: Global Sovereign Debt & Creditor Concentration Monitor

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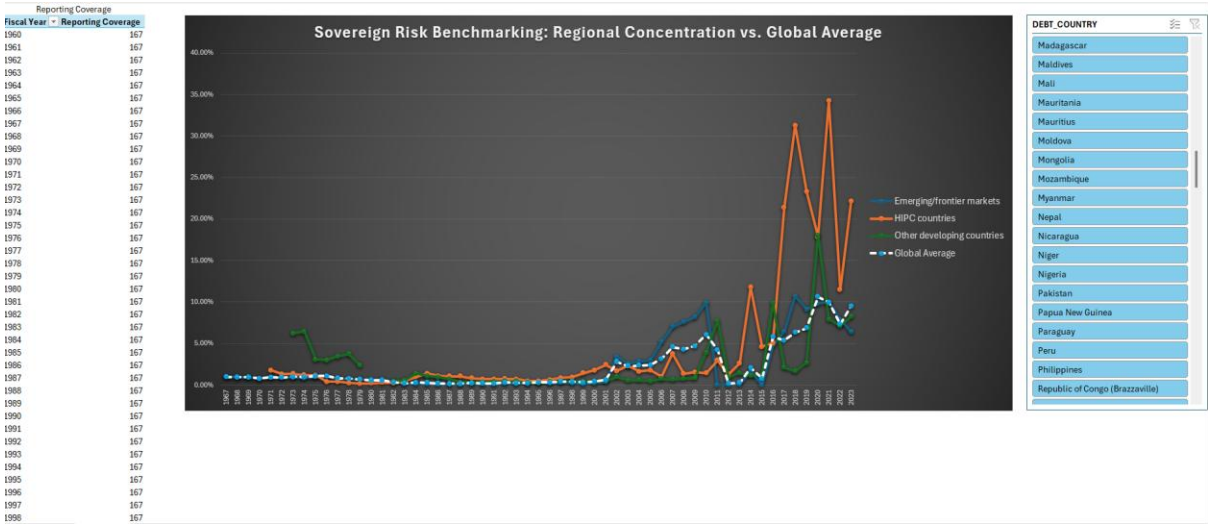
Domain Expertise: Financial Data Analysis | Risk Operations | Regulatory Compliance

Dataset: Bank of England / Bank of Canada (BoC–BoE) Sovereign Default Database (2024 Edition)

1. Project Objective & Business Value

The primary objective of this project was to engineer a high-performance analytical tool to monitor 60+ years of sovereign debt trends. While traditional reporting often focuses on total debt-to-GDP, this framework specifically isolates **Creditor Concentration Risk** to identify over-reliance on single lending blocks.

In the context of emerging markets, understanding the identity of the creditor is as vital as the amount of the debt. "Hidden" debt concentration poses a significant threat to audit transparency and sovereign creditworthiness. This tool provides a benchmarked view of regional debt governance against global standards to mitigate these risks.



2. Discussion: Insights from the Original Source

The project is built upon the **2024 BoC–BoE Sovereign Default Database**, a collaborative effort designed to systematically measure and aggregate the nominal value of various types of sovereign debt in default.

- **Strategic Mission:** The database supports the central banks’ mission to promote economic stability and financial well-being.
- **Addressing Information Gaps:** Before this database was established in 2014, few efforts had been made to aggregate such comprehensive estimates of government obligations.
- **The 2024 "Concentration" Reality:** The source material highlights that the distribution of defaults remains highly concentrated, with only 10 sovereigns accounting for 75% of the total US-dollar value of debt in default globally.
- **Historical Accuracy:** The 2024 update includes significant historical revisions, most notably for the repudiated debts of Imperial Russia (1917) and China (1949), which were later settled in the late 1980s and 1990s.

3. Technical Architecture (The Process)

To ensure the monitor is scalable and "audit-ready," a **Three-Tier Data Architecture** was implemented:

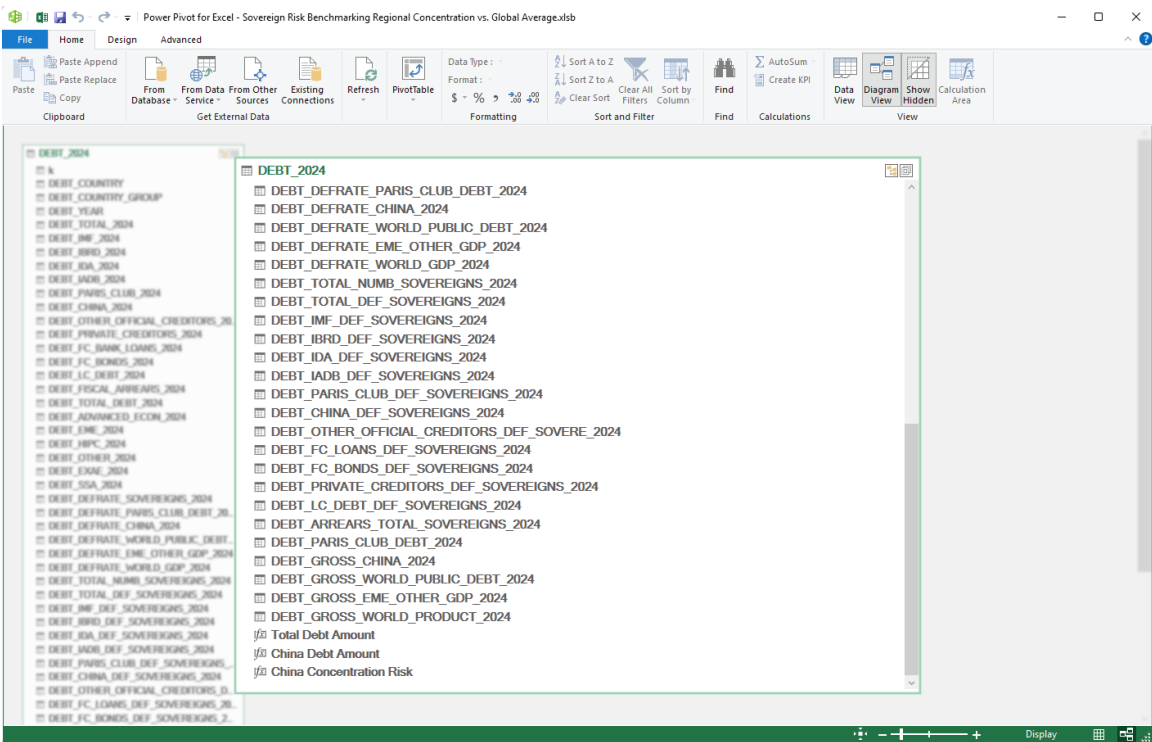
Tier 1: Data Engineering (Power Query ETL)

- **Ingestion:** Automated the connection to the BoE/BoC 2024 CSV, JSON, and XML data, managing complex multi-line metadata headers.
- **Normalization:** Transformed static historical records into a dynamic time-series format for 167 country profiles.

Tier 2: Analytical Modelling (Power Pivot & DAX)

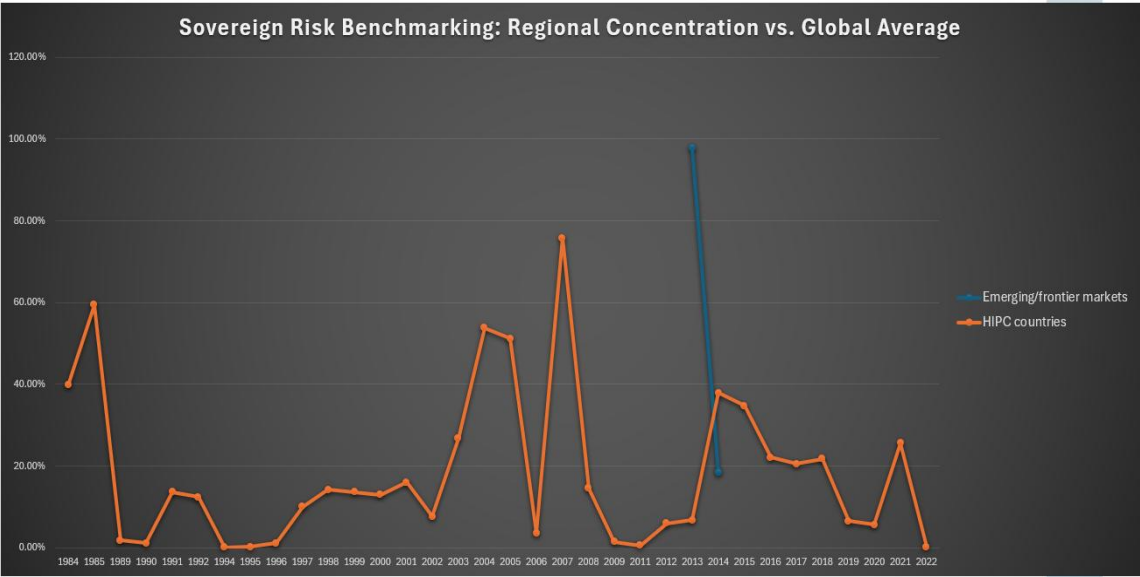
Custom DAX (Data Analysis Expressions) were authored to calculate core engine metrics:

- **Total Debt Volume:** $\text{=SUM}([\text{DEBT_TOTAL_2024}])$
- **Targeted Creditor Exposure:** $\text{=SUM}([\text{DEBT_CHINA_2024}])$
- **Concentration Risk Index:** $\text{=DIVIDE}([\text{China Debt Amount}], [\text{Total Debt Amount}])$



Tier 3: Executive Visualization (Presentation Layer)

- **Trend Analysis:** A Multi-Series Line Chart displays the Concentration Risk Index across various regions and the global average.
- **Interactivity:** Synchronized Slicers allow for instant toggling between regions and specific nations, such as Nigeria or Mali.



4. Key Findings: Global Trends & Regional Surges

The tool identified several critical shifts documented in the 2024 database update:

- **The "China Shift":** Identified defaults to China rose by 28% (nearly \$50 billion) in 2023, while defaults to the Paris Club remained steady at approximately \$78 billion.
- **Concentrated Value:** Just three sovereigns—Venezuela, Russia, and Iraq—accounted for 35% of the overall amount in default in 2023.
- **The 2021 HIPC Peak:** China’s concentration risk in Heavily Indebted Poor Countries (HIPC) peaked at 34.2% in 2021.
- **Nigeria Deep-Dive:** Analysis reveals localized spikes in concentration following major infrastructure financing cycles, particularly around 2013–2014.

5. Data Governance & Internal Controls

Reflecting a background in Operational Risk, the tool includes two specific internal controls to ensure data integrity:

- **Reporting Coverage Monitor:** A validation table ensures historical trends are not skewed by reporting lags (stabilized at ~167 countries).
- **Risk Threshold Alerts:** Conditional formatting visually flags any region exceeding a 20% concentration threshold for mandatory regulatory review.

Reporting Coverage	
Fiscal Year	Reporting Coverage
1960	167
1961	167
1962	167
1963	167
1964	167
1965	167
1966	167
1967	167
1968	167
1969	167
1970	167
1971	167
1972	167
1973	167
1974	167
1975	167

6. Conclusion

Technical Conclusion

This project demonstrates end-to-end mastery of the data lifecycle, transforming raw, multi-format central bank data into a scalable executive decision-making tool. By implementing a three-tier architecture and custom DAX modelling, the monitor ensures data integrity and provides the transparency required to mitigate Concentration Risk and maintain fiscal compliance.

Layman Conclusion

Think of a country like a person with several credit cards. While most banks only look at the total amount that person owes, this project looks at who they owe it to. If a country owes almost all its debt to one single lender, that lender has a lot of power over them, which is a big risk. This tool acts as an early warning system that flags when a country is becoming too dependent on one source like the recent 28% jump in debt owed to China helping experts catch financial problems before they turn into a global crisis.